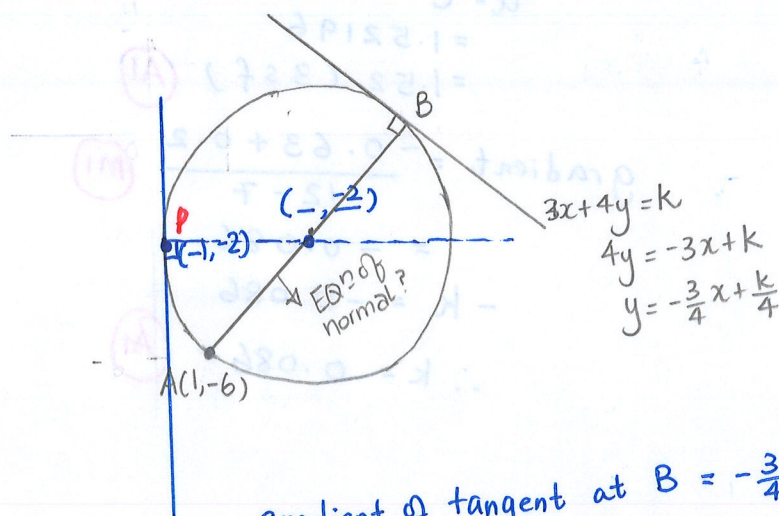


- 13 A circle has a diameter AB . The point A has coordinates $(1, -6)$ and the equation of the tangent to the circle at B is $3x + 4y = k$.

$$4y = -3x + k$$

- (a) Show that the equation of the normal to the circle at the point A is $4x - 3y = 22$. [3]



gradient of tangent at $B = -\frac{3}{4}$

gradient of normal at $B = \frac{4}{3} = \text{grad normal at } A$

Eqⁿ of normal at $A: A(1, -6)$

$$y - y_1 = m(x - x_1)$$

$$y + 6 = \frac{4}{3}(x - 1)$$

$$y = \frac{4}{3}x - \frac{4}{3} - 6$$

$$y = \frac{4}{3}x - \frac{22}{3}$$

$$\frac{4}{3}x - y = \frac{22}{3}$$

$$\times 3, \quad 4x - 3y = 22 \text{ (shown)}$$

- (b) It is also given that line $x = -1$ touches the circle at the point $(-1, -2)$.
Find the coordinates of the centre and the radius of the circle.
Hence, state the equation of the circle.

[5]

Since $x = -1$ touches the circle at pt $(-1, -2)$,

So equation of normal at $(-1, -2)$ is

$y = -2$ & this passes thro' the centre of circle

$$y = -2 \quad \text{--- (1)}$$

$$4x - 3y = 22 \quad \text{--- (2)}$$

subst. (1) into (2),

$$4x - 3(-2) = 22$$

$$4x = 22 - 6$$

$$4x = 16$$

$$x = 4$$

$\therefore$ centre of circle is $(4, -2)$

C $(4, -2)$, A $(1, -6)$

$$\text{Radius} = \sqrt{(4-1)^2 + (-2+6)^2}$$

$$= \sqrt{9 + 16}$$

$$= 5 \text{ units}$$

$\therefore$ EQⁿ of circle:

$$(x-4)^2 + (y+2)^2 = 25$$

(c) Find the value of k .

[2]

$$A(1, -6), C(4, -2), B(m, n)$$

$$\text{Mid-point of } AB, M\left(\frac{1+m}{2}, \frac{-6+n}{2}\right) = (4, -2)$$

$$\frac{1+m}{2} = 4$$

$$1+m=8$$

$$m=7$$

$$\frac{-6+n}{2} = -2$$

$$-6+n=-4$$

$$n=2$$

$$\therefore B(7, 2) \quad \text{(M)}$$

$$3x + 4y = k$$

$$\therefore k = 3(7) + 4(2)$$

$$= 29$$

$$\text{(A)}$$

- END OF PAPER -